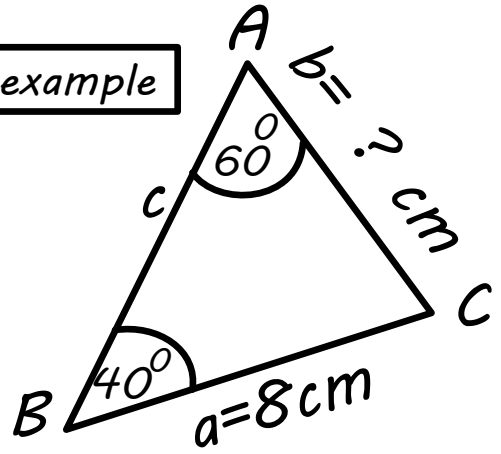
Law of Sines

Finding sides

For example



To use the law of sines we need to know…

1 side

2 angles

Now we can use the sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

what we know…

a=8cm A=60°

b= ? cm B=40°

For this example…

$$\frac{8}{\sin 60} = \frac{b}{\sin 40}$$

$$b = \frac{8 \sin 40}{\sin 60}$$

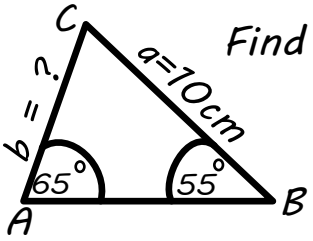
$$b = 5{\cdot}9378\ldots$$

b= _____(to 1 d·p)

remember

The angles in a triangle all add up to _____

Try it yourself



Find the length of b.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

b= _____ (to 1 d·p)

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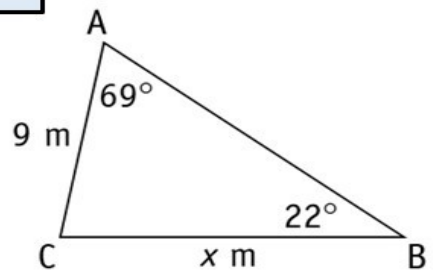
Lesson Objective: By the end of the lesson you should be very comfortable with using the Sine Rule and when to use it as well as how to rearrange when needed.

1



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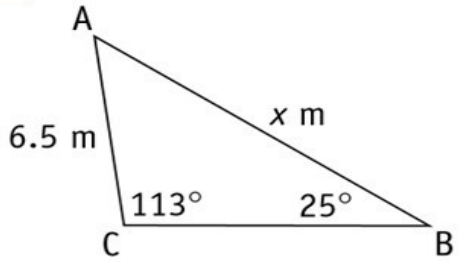

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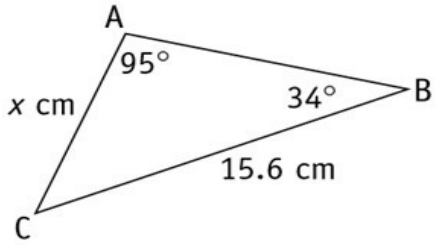

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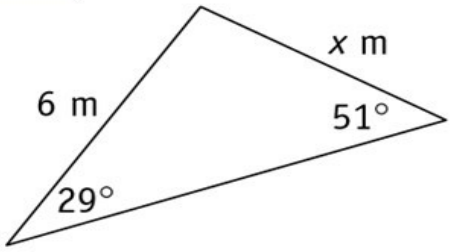

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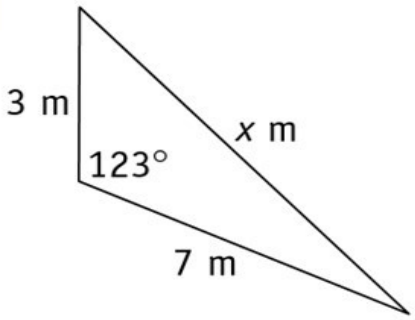

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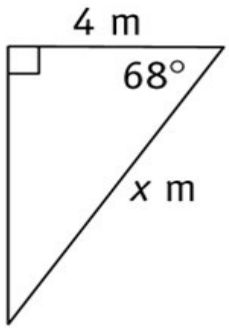

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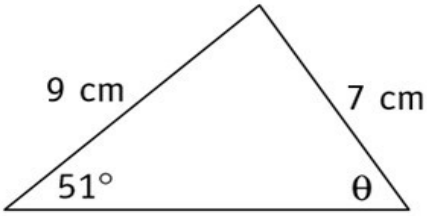

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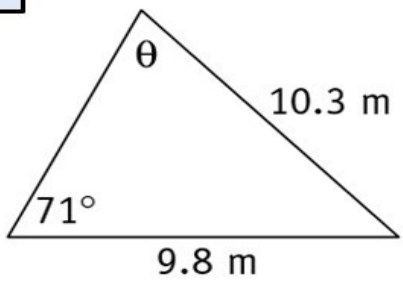

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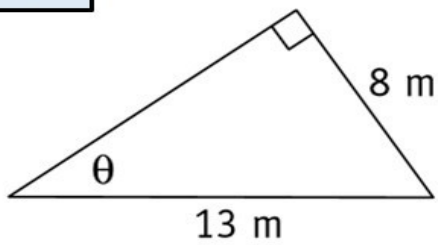

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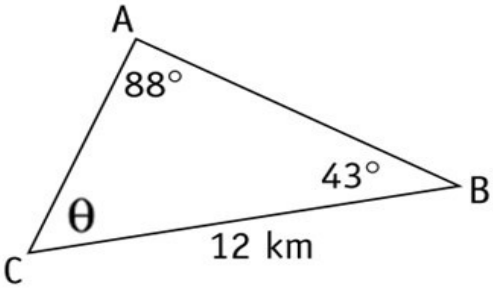

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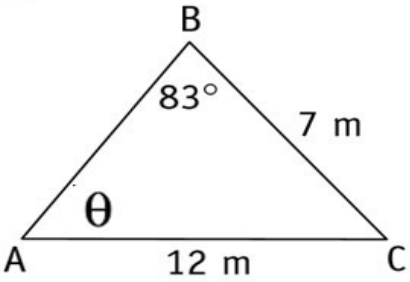

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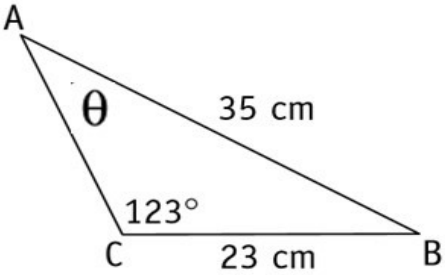

Self Evaluation: What have you learnt today? Are you confident with the sine rule? Did you struggle when finding the angle instead of a side? Or with rearranging?